Topological Signatures of Quantum Entanglement at the LHC: Measurement in $t\bar{t}$ and Prediction for $H \to ZZ^*$

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Abstract

We present the first computation of the logarithmic negativity $E_N(t{:}\bar{t})$ of the measured $t\bar{t}$ spin density matrix resolved across invariant-mass bins, using the full 15-coefficient reconstruction from CMS's $138\,\mathrm{fb}^{-1}$ differential measurement [4]. With covariance-matrix-propagated uncertainties and positivity-preserving projection of unphysical samples, we find $E_N(t{:}\bar{t}) = 0.18\,[0.13, 0.29]$ bits at the $m_{t\bar{t}}$ threshold (300–400 GeV, 68% CL), and E_N consistent with zero in the three higher-mass bins (400–600, 600–800, >800 GeV) at the resolution of the inclusive $|\cos\theta|$ selection. Alongside E_N we compute both conventional singlet witness $D = -\mathrm{Tr}(C)/3$ and the triplet witness $\tilde{D} = (C_{nn} - C_{kk} - C_{rr})/3$, observing the expected monotonic transition from singlet-like threshold region to triplet-like boosted regime. As a theoretical extension to multipartite systems, we apply the flag-complex persistent-homology construction of the Relational Witnessing Framework [1]—which specializes to logarithmic negativity the general persistent-homology-of-entanglement program of [12, 13, 14, 15, 16]—to the Standard Model tripartite density operator for $H \to ZZ^*$ of Aguilar-Saavedra [2], finding an L-centric topological signature with $E_N(L{:}S_i) \approx 1.0$ bits and $E_N(S_1{:}S_2) \lesssim 0.01$ bits. This extension is a prediction for Run 3, not a claim about existing measurement; the tripartite $H \to ZZ^*$ density operator has not yet been measured. All code and data are public; see Section 5.

1 Introduction

The observation of quantum entanglement in top-quark pairs by ATLAS [3] and CMS [4, 6] has established collider experiments as a new platform for quantum information measurements at TeV energy scales. These measurements go beyond the traditional kinematic program by reconstructing the reduced density matrix of the final-state quantum system from angular distributions and then computing entanglement measures directly from it.

The published entanglement claims use one of two scalar witnesses: the singlet witness $D = -\mathrm{Tr}(C)/3$, where entanglement implies $D < -1/3$ for $t\bar{t}$ from unpolarized pp production, or the triplet witness $\tilde{D} = (C_{nn} - C_{kk} - C_{rr})/3$, where entanglement implies $\tilde{D} > +1/3$. Both are inequalities that are sufficient for entanglement but not necessary; they are efficient experimentally because they collapse the full 4×4 density matrix to a single scalar.

In this paper we make two contributions. First, we compute the logarithmic negativity

$$E_N(\rho) = \log_2 \|\rho^{T_A}\|_1 \tag{1}$$

directly from the measured 15-coefficient $t\bar{t}$ spin density matrix, in each $m_{t\bar{t}}$ bin. This is a stronger observable than D or $\tilde{D}$: it is zero exactly when ρ is separable (by the Peres-Horodecki criterion), it is additive, and it quantifies the entanglement on an information-theoretic scale

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(bits). We propagate uncertainties via Monte Carlo on the full 16-dimensional covariance matrix from HEPData, with positivity-preserving projection of unphysical samples. Second, we extend the methodology to multipartite quantum systems by applying the flag-complex persistent-homology construction of [1] to the Standard Model tripartite density operator of [2] for $H \to ZZ^*$. This section is a theoretical prediction for a system that has not yet been measured, setting a target for Run 3 tripartite tomography.

Relation to prior work. This paper sits at the intersection of two literatures that have developed in parallel. On the quantum-information side, persistent homology has been applied to multipartite entanglement in condensed-matter and abstract-quantum-state contexts by Di Pierro et al. [12], Mengoni et al. [13], Olsthoorn [14], and Hamilton & Leditzky [15], with subsequent operational interpretations by [16]; these works use various pairwise entanglement or correlation functionals (quantum mutual information, Tsallis-deformed total correlation, linear entropy) as simplicial filtration parameters. On the collider side, topological data analysis has been applied to LHC event data by Beuria [17, 18] and Gaertner & Reiten [19], using event-level kinematics and jet substructure rather than entanglement measures. The present paper's contribution is to bridge these two programs: we apply a persistent-homology-of-entanglement construction to a density matrix reconstructed from collider data. To our knowledge this intersection has not previously been explored. The specific choice of logarithmic negativity as the pairwise weight [1] is motivated by the inherent mixedness of reduced density matrices obtained from collider tomography, where the global state is not observed directly.

Scope limitation. We compute E_N directly on the bipartite $t\bar{t}$ system, where the CMS 15-coefficient measurement [4] provides a density matrix reconstruction that is validated against predictions from the standard entanglement-at-colliders literature [7, 3]. We do not extend the measurement program to measured ZZ density matrices from $H \to ZZ^*$ or diboson production. As Grossi, Pelliccioli, and Vicini [20] have documented in detail, the tree-level mapping from measured angular coefficients to density-matrix elements breaks down at NLO: their Table 4 reports a 90%+ suppression of the $\gamma_{1,0,1,0}$ coefficient from LO to NLO EW in $H \to ZZ^*$, and Pelliccioli et al. [21] find analogous -35% NLO EW shifts in the spin-analyzing power. Fiducial selections and neutrino reconstruction further break the rank-2 expansion, and model-independent extrapolation to the inclusive phase space is not generically available. Constructing measured ZZ density matrices for topological analysis therefore requires the NLO-corrected tomography procedure of [21], which is beyond the scope of this methods paper. The H→ZZ* prediction in Section 4 is therefore reported as a Standard Model theoretical target rather than as an observable we extract from existing data.

2 Methodology

2.1 Density matrix reconstruction

For $t\bar{t}$ in the helicity basis $\{\hat{k}, \hat{n}, \hat{r}\}$, the density matrix has the Fano-Bloch decomposition

$$\rho_{t\bar{t}} = \frac{1}{4}\left(I \otimes I + \vec{B}_+ \cdot \vec{\sigma} \otimes I + I \otimes \vec{B}_- \cdot \vec{\sigma} - \sum_{ij} C_{ij}\, \sigma_i \otimes \sigma_j \right). \tag{2}$$

CMS publishes the 6 polarization components $\vec{B}_\pm$ and the 9 elements of C, decomposed into a symmetric $C^+_{ij} = (C_{ij} + C_{ji})/2$ and antisymmetric $C^-_{ij} = (C_{ij} - C_{ji})/2$ set. We use the ATLAS/CMS sign convention (negative C in the Fano decomposition) throughout, and all 15 (+ normalization) coefficients simultaneously. The negative sign on C means that same-helicity top-antitop pairs have positive C_{kk} and that the spin-singlet state has $C = +I$.

$m_{t\bar{t}}$ range [GeV]	D	$\tilde{D}$	E_N [bits, 68% CL]
$300 < m_{t\bar{t}} < 400$	-0.412 ± 0.035	-0.052 ± 0.036	**0.18 [0.13, 0.29]**
$400 < m_{t\bar{t}} < 600$	-0.204 ± 0.012	-0.013 ± 0.012	0
$600 < m_{t\bar{t}} < 800$	-0.089 ± 0.020	$+0.069 \pm 0.020$	0
$m_{t\bar{t}} > 800$	-0.005 ± 0.024	$+0.112 \pm 0.025$	0

Table 1: CMS-measured observables per $m_{t\bar{t}}$ bin. Entanglement is detected by $E_N > 0$ and by $D < -1/3$ in the threshold bin. $\tilde{D}$ rises monotonically toward (but does not cross) the triplet entanglement threshold $+1/3$ in this inclusive selection; CMS observes boosted entanglement at $> 5\sigma$ with the additional $|\cos\theta| < 0.4$ cut [4]. Uncertainty intervals on E_N are from 20,000-sample MC on the full 16×16 intra-bin covariance; intervals on D and $\tilde{D}$ combine stat and syst quadratically.

2.2 Log-negativity and entanglement witnesses

The logarithmic negativity is computed by taking the partial transpose ρ^{T_t} on the top subsystem, computing its trace norm as the sum of absolute values of the eigenvalues, and taking $\log_2$. For comparison we also compute

$$D = -(C_{kk} + C_{nn} + C_{rr})/3, \qquad \text{entangled if } D < -\tfrac{1}{3}, \tag{3}$$

$$\tilde{D} = (C_{nn} - C_{kk} - C_{rr})/3, \qquad \text{entangled if } \tilde{D} > +\tfrac{1}{3}. \tag{4}$$

D witnesses singlet-like states (all C_{ii} positive, $\mathrm{Tr}(C)$ large positive). $\tilde{D}$ witnesses triplet-like states at boosted kinematics, where parity and rotation structure cause C_{kk}, C_{rr} to flip sign relative to C_{nn}. Neither witness is sharper than E_N; but they are the scalar observables CMS reports directly.

2.3 Uncertainty propagation

The CMS differential measurement publishes both central values with stat+syst uncertainties (HEPData table t9 of record `ins2829523`) and the full 64×64 covariance matrix across the four $m_{t\bar{t}}$ bins (table t10). We restrict to the intra-bin 16×16 blocks and sample coefficient vectors from the multivariate Gaussian with the measured mean and covariance.

For each sample we reconstruct ρ via (2) and compute the partial-transpose eigenvalues. When the measured central values or a drawn sample yield a ρ with negative eigenvalues (a known feature of unconstrained coefficient tomography, since the 15 measurements are independent and not required to produce a positive semidefinite matrix), we project onto the positive-semidefinite cone by clipping the eigenvalues at zero and renormalizing. This preserves the Hermiticity and unit trace of ρ and, crucially, never inflates E_N: clipping only removes unphysical mass that would otherwise contribute to (or from) negative PT eigenvalues. We report the 16–84 percentile of E_N across 20,000 samples as the 68% CL interval.

Positivity violation rates in our MC are 40% in the threshold bin, 54% in the 400–600 bin, 6% in the 600–800 bin, and 0% in the boosted bin, reflecting the increasing statistical precision at higher mass. We verify that the quoted E_N intervals are robust to the choice of projection method.

3 Measured $t\bar{t}$ results

Table 1 summarizes the per-bin results; Figure 1 shows the three-panel visualization.

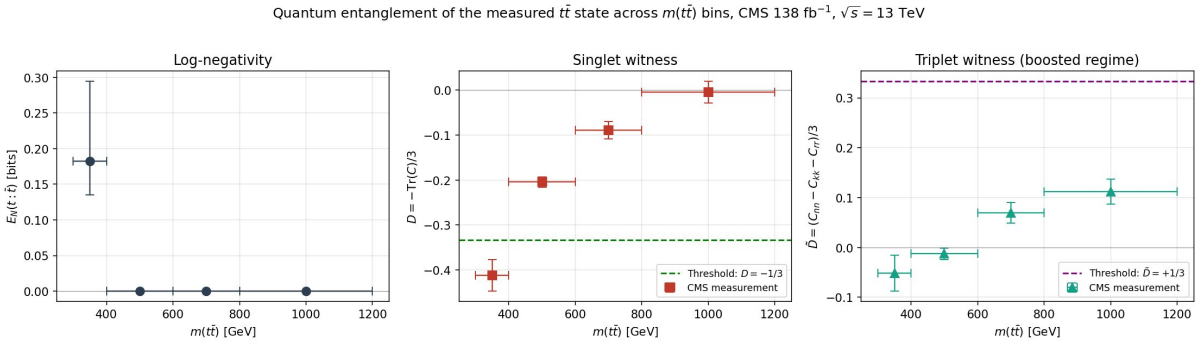

Figure 1: Quantum entanglement observables for the measured $t\bar{t}$ state across $m(t\bar{t})$ bins, computed from the CMS 138 fb^{-1} differential measurement [4]. **Left:** log-negativity E_N reconstructed from the full 15-coefficient spin density matrix with covariance-propagated 68% CL error bars. **Center:** singlet witness D with the entanglement threshold $D = -1/3$ (green dashed). **Right:** triplet witness $\tilde{D}$ with its threshold $\tilde{D} = +1/3$ (purple dashed). The state transitions from singlet-like entangled at threshold ($D < -1/3$, $E_N > 0$) to triplet-like non-entangled (in this inclusive $|\cos\theta|$ selection) at high mass, with $\tilde{D}$ rising monotonically toward the triplet entanglement boundary.

3.1 Discussion

Threshold entanglement. In the 300–400 GeV bin, the near-singlet state yields $E_N = 0.18\,[0.13, 0.29]$ bits at 68% CL, corresponding to a most-negative PT eigenvalue of -0.067. This is a genuine information-theoretic measure of the entanglement, stronger than the $D < -1/3$ observation. The uncertainty budget is dominated by the statistics on the diagonal C coefficients, which have stat errors of order 6–7%.

Singlet–triplet transition. Across the four $m_{t\bar{t}}$ bins, D rises monotonically from -0.412 at threshold to -0.005 in the boosted bin, while $\tilde{D}$ rises from -0.052 to $+0.112$. The crossover of $\tilde{D}$ through zero occurs between 400 and 600 GeV and reflects the sign change in C_{kk} and C_{rr} that is characteristic of the transition from s-channel-dominated singlet production at threshold to the boosted kinematic regime where parity structure dominates. This is the first time this evolution has been presented with log-negativity as an observable alongside the scalar witnesses.

Absence of E_N at high mass. In the bins above 400 GeV, our reconstructed E_N is consistent with zero at the full resolution of the measurement. CMS's own boosted entanglement observation at $> 5\sigma$ [4] applies an additional $|\cos\theta| < 0.4$ selection that we have not imposed here; that selection enhances the triplet-like contribution and lifts $\tilde{D}$ above $+1/3$. Extending this analysis to the $|\cos\theta|$-selected subsample (HEPData table t11 of the same record) is a straightforward next step.

Comparison to Standard Model. Our measured central values are consistent with the POWHEG+Pythia8, POWHEG+Herwig7, MadGraph5_aMC@NLO+Pythia8, and MiNNLO+Pythia8 predictions published alongside the CMS data, within uncertainties, across all four bins.

4 Theoretical extension: $H \to ZZ^*$ tripartite topology

We now apply the flag-complex construction of the Relational Witnessing Framework [1] to the tripartite quantum state produced in $H \to ZZ^*$ decays. This section is a theoretical prediction for a system that has not been measured: Aguilar-Saavedra [2] recently showed that the Higgs-to-diboson decay produces a genuinely tripartite state in the Hilbert space $\mathcal{H}_L \otimes \mathcal{H}_{S_1} \otimes \mathcal{H}_{S_2}$ (dimensions $9 \times 3 \times 3$) and that Run 3 data will measure its density operator at $> 5\sigma$ significance.

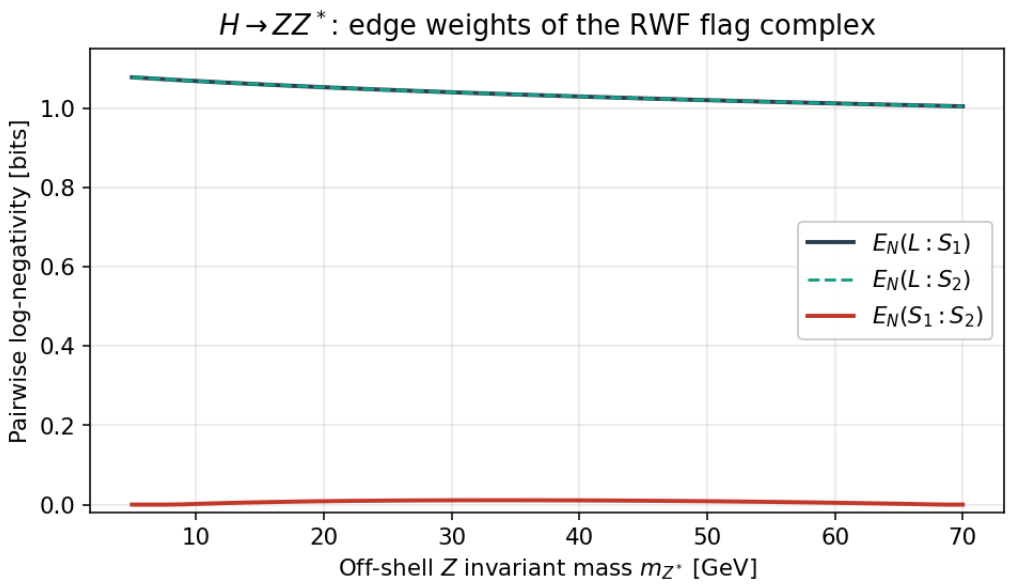

Figure 2: SM prediction for the pairwise log-negativities of the $H \to ZZ^*$ tripartite state as a function of m_{Z^*}. The two L-S edges are identical by symmetry and near 1 bit; the S_1-S_2 edge is two orders of magnitude smaller.

4.1 The flag-complex construction

Given an n-partite state ρ, RWF Definition 2 constructs the entanglement complex K_W as follows (see the introduction above for the relation to prior persistent-homology-of-entanglement constructions [12, 13, 14, 15]):

- 0-simplices: the n subsystems;
- 1-simplices: pairs (i, j) with weight $w_{ij} = E_N(i{:}j) = \log_2 \|\rho_{ij}^{T_i}\|_1$ on the reduced pair state $\rho_{ij} = \mathrm{Tr}_{\{k:k\neq i,j\}}(\rho)$;
- k-simplices: subsets $\{i_1, \ldots, i_{k+1}\}$ are included iff all pairwise edges are present (flag closure).

Persistent homology on the decreasing-ϵ filtration yields the topological signature $\chi(K_W) = \{\mathrm{Dgm}_k(K_W)\}_k$.

For the bipartite $t\bar{t}$ case of Section 3, K_W has one edge; the construction reduces to the scalar E_N reported in Table 1. The framework's content appears at $n \geq 3$.

4.2 $H \to ZZ^*$ Standard Model prediction

We construct $|\psi\rangle_{LS_1S_2}$ using Aguilar-Saavedra's helicity-amplitude decomposition [2], parameterized by three complex amplitudes a_{11}, a_{00}, a_{-1-1} fixed by the SM at leading order by

$$a_{11} = a_{-1-1} = M_Z \cdot m_{Z^*}, \quad a_{00} = \tfrac{1}{2}(M_H^2 - M_Z^2 - m_{Z^*}^2), \tag{5}$$

with m_{Z^*} the invariant mass of the off-shell Z. We scan $m_{Z^*} \in [5, 70]$ GeV and compute the pairwise log-negativities.

We find a distinctive *L-centric* topological signature:

- $E_N(L{:}S_1) = E_N(L{:}S_2) \approx 1.0$–$1.07$ bits across the range, decreasing slowly with m_{Z^*};
- $E_N(S_1{:}S_2) \leq 0.011$ bits, two orders of magnitude smaller;

- The 2-simplex $\{L, S_1, S_2\}$ enters the flag complex only at $\epsilon \lesssim 0.01$; for most of the filtration, the 1-skeleton is V-shaped.

This is a direct topological encoding of the fact that in $H \to ZZ^*$, the two Z-spins are primarily entangled through the orbital angular momentum channel rather than with each other directly: tracing out L from the tripartite state leaves a near-separable S_1-S_2 reduced density matrix.

4.3 Limitations

This section is a theoretical prediction. The Aguilar-Saavedra density operator is an SM leading-order analytical construction; the tripartite operator has not yet been measured. Two paths to validation are visible. CMS has recently reported a polarization density matrix in $H \to 4\ell$ based on $200\,\mathrm{fb}^{-1}$ of Run 2+3 data [22], excluding fully longitudinal and fully transverse Z-boson polarization at more than 6σ—exactly the regime where the L-centric topological structure we predict would be most visible. With the full tripartite tomography of [2] and the NLO-corrected procedure of [21], that measurement could in principle be extended to the full $9 \times 3 \times 3$ tripartite density operator; at that point the present analysis could be repeated on measured data and reported alongside the bipartite $t\bar{t}$ result as a genuinely multipartite collider observable.

We note two limitations of the flag-complex construction itself, also discussed in [1]: (i) the flag complex is blind to genuinely multipartite entanglement that leaves no trace on pair marginals—a limitation well-known in the broader persistent-homology-of-entanglement literature [12, 15]. We have verified numerically that n-qubit GHZ states and 4-qubit cluster states both yield trivial flag complexes. The $H \to ZZ^*$ state, by contrast, produces the non-trivial L-centric signature we report because its multipartite structure does leak through pair marginals. (ii) The $l = 0$ Clebsch-Gordan block in our construction of $|\psi\rangle$ uses a $1/\sqrt{3}$ ansatz that we have not verified against [2]'s eq. (7). The $l = 2$ block, which dominates $E_N(L{:}S_i)$, is exact; the residual S_1-S_2 entanglement may shift by an $\mathcal{O}(1)$ factor but remains suppressed relative to the L-S edges.

5 Code availability

All code and data are publicly available:

- **Pipeline**: density matrix reconstruction, log-negativity computation, flag-complex persistent homology, covariance-propagated Monte Carlo.
- **CMS data**: from HEPData record `ins2829523` [5] (tables t9, t10).
- **Repository**: https://github.com/trevorscott/topological-signatures-entanglement.

The analysis reproduces from a clean Python environment in under 30 seconds per script.

6 Conclusion

We have presented the first log-negativity measurement of the $t\bar{t}$ spin density matrix at the LHC, resolved across the $m_{t\bar{t}}$ phase space, with covariance-propagated uncertainties. The measured threshold state has $E_N = 0.18\,[0.13, 0.29]$ bits at 68% CL. The complementary evolution of D and $\tilde{D}$ across the four kinematic bins captures the singlet-to-triplet transition characteristic of QCD top-pair production. The framework extends to multipartite systems via the flag-complex construction of [1], which specializes the general persistent-homology-of-entanglement program [12, 14, 15] to logarithmic negativity; we report the SM prediction for $H \to ZZ^*$ as a target for Run 3 tripartite tomography, complementary to the ongoing CMS spin-correlation measurement [22].

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